Answer Key & Teacher Notes

Session 1 — Surface Area of Rectangular Prisms · Session 2 — Surface Area of Triangular Prisms

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Surface Area of Rectangular Prisms

TALK PROMPTS

- Why is surface area measured in square units even though the chest is a solid?
- Which two edges make each face? Show your partner on the net.
- Why must you round up when you count cans of stain?

PARTNER TASK

Gift Wrap. Partners each find the three different face areas of a box 12 in by 10 in by 7 in (120, 84, 70 in²). One partner doubles and adds; the other checks by labeling every face on a net. Both should get 548 in².

WATCH FOR

- Multiplying l × w × h (volume) when surface area is asked.
- Adding the three different faces without doubling them.
- Rounding cans of paint down, or including a face that is not covered.

SESSION 2 · TEACHER NOTES

Surface Area of Triangular Prisms

TALK PROMPTS

- Why do all three rectangles in this net share the same length?
- How do you decide which measure is the triangle's height and which is just a side?
- Where does the 6 cm show up in each of the cheese wedge's three rectangles?

PARTNER TASK

Doorstop Check. Partner A finds both triangle ends of a prism 8 cm long with right-triangle ends 6, 8, 10 cm (48 cm²); Partner B finds the three rectangles (192 cm²). Add for 240 cm², then each explains which number was used as a height and which as a side.

WATCH FOR

- Using the triangle's height as the side of a rectangle.
- Counting only one triangle end.
- Forgetting the ½ in the triangle area.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.4	1	2	1	1	1	2	1	3	2	1	1
S1 · Form B	6.G.A.4	1	2	1	1	1	2	1	3	2	1	1
S2 · Form A	6.G.A.4	1	2	2	1	2	2	2	3	2	1	1
S2 · Form B	6.G.A.4	1	2	2	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

surface area: The total **area** of all the faces of a solid.

congruent: Exactly the same **size** and shape.

net: A flat **pattern** that folds into a solid.

- A rectangular prism has **3** pairs of matching faces.
- SA = 2lw + 2lh + **2wh**.
- Surface area is measured in **square** units.

Example: Top: 6 × 4 = **24** cm². Front: 6 × 3 = 18 cm². End: 4 × 3 = **12** cm². Double each: 48 + 36 + 24. Answer: SA = **108** cm²

YOU TRY

1. 2(5 × 2) + 2(5 × 3) + 2(2 × 3) = 20 + 30 + 12 = 62 in²

2. a. 6 × 4 = 24 m² b. 6 × 100 = 600 in²

EXIT TICKET

1. a. 6 × 9 = 54 ft² b. 2(12) + 2(8) + 2(6) = 52 cm²

2. 2(8 × 5) + 2(8 × 2) + 2(5 × 2) = 80 + 32 + 20 = 132 in²

3. **B** (2lw + 2lh + 2wh)

PRACTICE FORM A

1. a. 6 × 64 = 384 cm² b. 2(54) + 2(36) + 2(24) = 228 cm² c. 2(50) + 2(30) + 2(15) = 190 in² d. 6 × 36 = 216 m²

2. a. top 15 ft², front 10 ft², end 6 ft² b. 2(15) + 2(10) + 2(6) = 62 ft² c. 62 ÷ 20 = 3.1, so 4 cans (round up)

3. Faces 120, 84, 70 in²; SA = 2(120) + 2(84) + 2(70) = 548 in²

4. a. 6 × 25 = 150 cm² b. 6 × 49 = 294 cm²

5. 2(48) + 2(24) + 2(18) = 180 ft²

6. No. 144 in² is one face. A cube has 6 faces: 6 × 144 = 864 in².

7. Faces 180, 150, 120 in²; SA = 360 + 300 + 240 = 900 in²

8. a. 184 ft² b. 184 ÷ 40 = 4.6, so 5 cans c. 184 − 40 = 144 ft²; 3.6, so 4 cans

9. Jonah found the volume. Surface area adds the areas of the 6 faces: 2(70) + 2(50) + 2(35) = 310 cm².

10. **C** (96 cm²)

11. **B** (3)

Self-check word **SURFACES**

PRACTICE FORM B

1. a. 6 × 81 = 486 cm² b. 2(40) + 2(24) + 2(15) = 158 cm² c. 2(48) + 2(24) + 2(8) = 160 in² d. 6 × 25 = 150 m²

2. a. top 24 ft², front 18 ft², end 12 ft² b. 2(24) + 2(18) + 2(12) = 108 ft² c. 108 ÷ 25 = 4.32, so 5 cans (round up)

3. Faces 126, 84, 54 in²; SA = 2(126) + 2(84) + 2(54) = 528 in²

4. a. 6 × 36 = 216 cm² b. 6 × 64 = 384 cm²

5. 2(63) + 2(36) + 2(28) = 254 ft²

6. No. 121 in² is one face. A cube has 6 faces: 6 × 121 = 726 in².

7. Faces 160, 128, 80 in²; SA = 320 + 256 + 160 = 736 in²

8. a. 280 ft² b. 280 ÷ 50 = 5.6, so 6 cans c. 280 − 60 = 220 ft²; 4.4, so 5 cans

9. Kara found the volume. Surface area adds the areas of the 6 faces: 2(54) + 2(45) + 2(30) = 258 cm².

10. **C** (76 m²)

11. **B** (square units)

Self-check word **PACKAGES**

GUIDED NOTES

triangular prism: A solid with 2 triangle **bases** and 3 rectangle faces.

lateral face: A side face that is **not** a base.

surface area: The total area of all the **faces**.

- Add the areas of all **5** faces: 2 triangles and 3 rectangles.
- Each triangle uses ½ × base × **height**.
- Each rectangle is one triangle **side** × the prism's length.

Example: One end: ½ × 8 × 3 = **12** ft². Both ends: 24 ft². Rectangles: 8 × 6 + 5 × 6 + 5 × 6 = **108** ft². Answer: SA = 24 + 108 = **132** ft²

YOU TRY

1. Ends 2 × (½ × 6 × 4) = 24 cm²; rectangles (6 + 5 + 5) × 7 = 112 cm²; SA = 136 cm²

2. a. 2 × 10 = 20 in² b. 36 in²

EXIT TICKET

1. a. 2 × 6 = 12 m² b. 12 × 2 = 24 m²

2. Ends 2 × (½ × 10 × 12) = 120 in²; rectangles (10 + 13 + 13) × 5 = 180 in²; SA = 300 in²

3. **D** (the triangle's height)

PRACTICE FORM A

1. a. 2 × 12 = 24 ft² b. 50 ft² c. 18 cm² d. 12 × 11 = 132 cm²

2. a. 2 × (½ × 24 × 5) = 120 cm² b. 24 × 20 + 13 × 20 + 13 × 20 = 1,000 cm² c. 120 + 1,000 = 1,120 cm²

3. Ends 2 × (½ × 20 × 21) = 420 cm²; rectangles (20 + 21 + 29) × 14 = 980 cm²; SA = 1,400 cm²

4. Ends 60 cm²; rectangles 30 × 6 = 180 cm²; SA = 240 cm²

5. No. 168 in² is one end. Both ends: 2 × 168 = 336 in².

6. Ends 2 × (½ × 36 × 24) = 864 ft²; rectangles (36 + 30 + 30) × 25 = 2,400 ft²; SA = 3,264 ft²

7. Ends 2 × (½ × 16 × 6) = 96 ft²; rectangles (16 + 10 + 10) × 10 = 360 ft²; SA = 456 ft²

8. a. 2 × (½ × 8 × 3) = 24 ft² b. 24 + 80 + 2 × (5 × 10) = 204 ft² c. 204 − 80 = 124 ft²

9. Amara counted only one triangle end. A prism has 2 matching ends: 2 × 24 = 48 cm². SA = 48 + 240 = 288 cm².

10. **B** (3)

11. **B** (prism's length)

Self-check word **TRIANGLE**

PRACTICE FORM B

1. a. 2 × 30 = 60 ft² b. 84 ft² c. 30 cm² d. 24 × 5 = 120 cm²

2. a. 2 × (½ × 16 × 6) = 96 cm² b. 16 × 15 + 10 × 15 + 10 × 15 = 540 cm² c. 96 + 540 = 636 cm²

3. Ends 2 × (½ × 8 × 15) = 120 cm²; rectangles (8 + 15 + 17) × 10 = 400 cm²; SA = 520 cm²

4. Ends 48 cm²; rectangles 24 × 5 = 120 cm²; SA = 168 cm²

5. No. 60 in² is one end. Both ends: 2 × 60 = 120 in².

6. Ends 2 × (½ × 24 × 16) = 384 ft²; rectangles (24 + 20 + 20) × 20 = 1,280 ft²; SA = 1,664 ft²

7. Ends 2 × (½ × 12 × 8) = 96 ft²; rectangles (12 + 10 + 10) × 12 = 384 ft²; SA = 480 ft²

8. a. 2 × (½ × 6 × 4) = 24 ft² b. 24 + 72 + 2 × (5 × 12) = 216 ft² c. 216 − 72 = 144 ft²

9. Felix used the triangle's height (4 ft) as a side. The rectangles use the triangle's sides: 6 × 10 + 5 × 10 + 5 × 10 = 160 ft². Ends 2 × (½ × 6 × 4) = 24 ft². SA = 184 ft².

10. **B** (5)

11. **B** (prism's length)

Self-check word **COVERING**

SHEET 1 · SURFACE AREA OF RECTANGULAR PRISMS

1. $2(10 \times 4) + 2(10 \times 5) + 2(4 \times 5) = 80 + 100 + 40 = 220$ cm²
2. a. 54 cm² b. 600 ft² c. 28 m² d. 90 in² e. $6 \times 25 = 150$ in² f. $2(18) + 2(12) + 2(6) = 72$ cm²
3. SA = $2(96) + 2(48) + 2(32) = 352$ in². Yes, 400 > 352, with 48 in² to spare.
4. Top $4 \times 2 = 8$; front and back $2 \times (4 \times 3) = 24$; sides $2 \times (2 \times 3) = 12$; $8 + 24 + 12 = 44$ ft²
5. He found the volume (8 cm³). Surface area adds the 6 faces: $6 \times (2 \times 2) = 24$ cm².

SHEET 2 · SURFACE AREA OF TRIANGULAR PRISMS

1. Triangles $2 \times (\frac{1}{2} \times 6 \times 4) = 24$; rectangles $6 \times 10 + 5 \times 10 + 5 \times 10 = 160$; SA = 184 m²
2. a. $2 \times (\frac{1}{2} \times 8 \times 3) = 24$ ft² b. 45 ft²
3. Ends 24; floor $8 \times 9 = 72$; sides $2 \times (5 \times 9) = 90$; total 186 ft²
4. Ends $2 \times (\frac{1}{2} \times 5 \times 12) = 60$; rectangles $5 \times 4 + 12 \times 4 + 13 \times 4 = 120$; SA = 180 in²
5. He used the triangle's height (4 m) as a side. The rectangles use the sides of the triangle: 6×10, 5×10, and 5×10.
6. **C** (82 cm²)
7. **B** (192 cm²)

October 2026 conceptual additions — teacher key

Session 1 · Surface Area of Rectangular Prisms

Closed: 2(15+10+6)=62 cm². Open: 62−15=47 cm². Count every included face exactly once.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Closed: 2(15+10+6)=62 cm². Open: 62−15=47 cm². Count every included face exactly once.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Surface Area of Triangular Prisms

Two ends: 2 × 1/2 × 3 × 4 = 12. Rectangles: (3+4+5) × 6 = 72. Total 84 cm². The 5 cm side is not the perpendicular height.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Two ends: 2 × 1/2 × 3 × 4 = 12. Rectangles: (3+4+5) × 6 = 72. Total 84 cm². The 5 cm side is not the perpendicular height.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.